



DEALER BEST PRACTICE SERIES

Safety Processes For A Mine Site

Application Maintenance Site Management Component Rebuild Component Life Management **Safety** MARC Management

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1.0 Introduction

Safety is an extremely important topic to companies that operate in and around the mining industry. There are a variety of safety procedures and policies in use around the world at mining companies and mining dealers. The procedures documented here were taken from a mining customer site that had reported success in reducing the number of reportable incidents. It is advised to the reader that there are extensive variations to the procedures documented here in use around the world and that this summary is offered as one particular example that has worked for a mining company. As additional examples of procedures are documented from other mining companies, it is anticipated they will be added here.

Reviewing current fatality and incident data clearly establishes the need for a sound safety policy at a mine site. Readily available data comes from the U.S. government's Mine Safety and Health Administration (MSHA) website. MSHA collects and publishes incident related data from mine sites throughout the U.S. The data revealed here is from 2005.



Figure 1¹

Figure 1 shows the trend in US mine site fatalities from 1995 to 2005. Although the trend is down, the ultimate goal is zero fatalities.

Figure 2 delineates the 1995 to 2005 fatality data. The chart shows percentages of each fatality in a particular classification. Leading by far, powered haulage ranked highest with 46% of all associated fatalities. Machinery was associated with 29% of all fatalities. Then following machinery, two classification groups were tied both with 8%; falling objects and falling persons.

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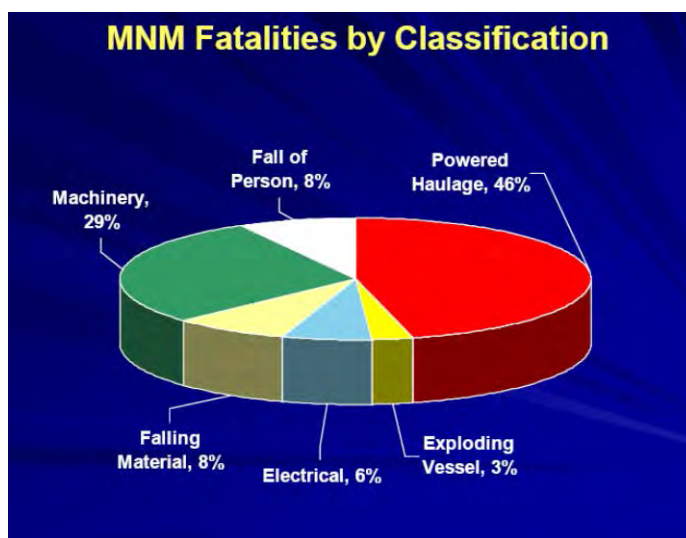
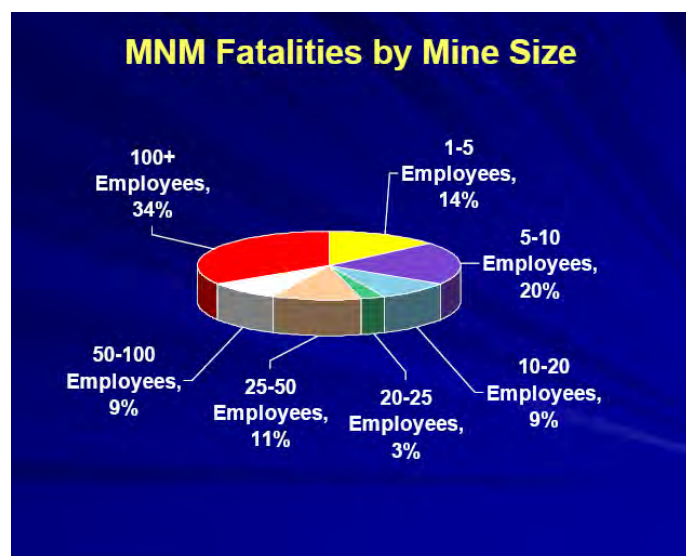
Figure 2¹Figure 3¹

Figure 3 classifies the same fatalities by size of mining operation. By far, the majority with 34% of all fatalities was mining sites that have more than 100 employees on site.

Figures 4 and 5 provide age and experience demographics associated with fatalities. Figure 3 shows that age is not clearly correlated to the root cause of a fatality. However, years of experience, shown in Figure 4, does show a correlation to the percentage of fatalities.

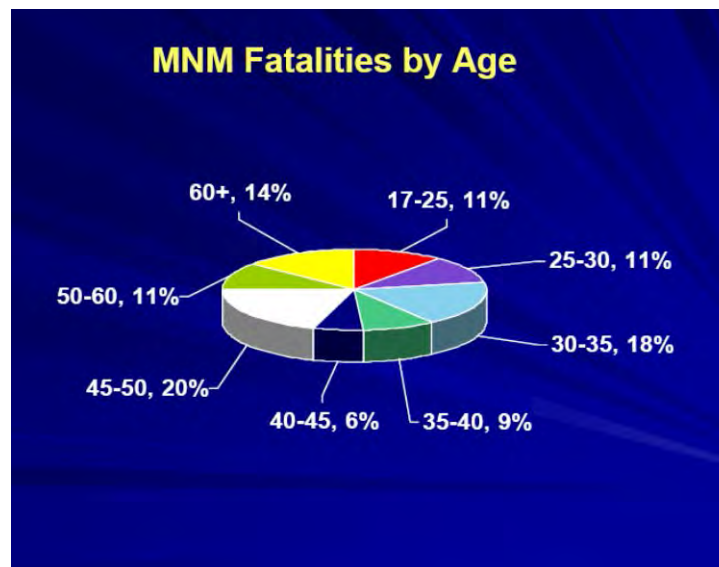
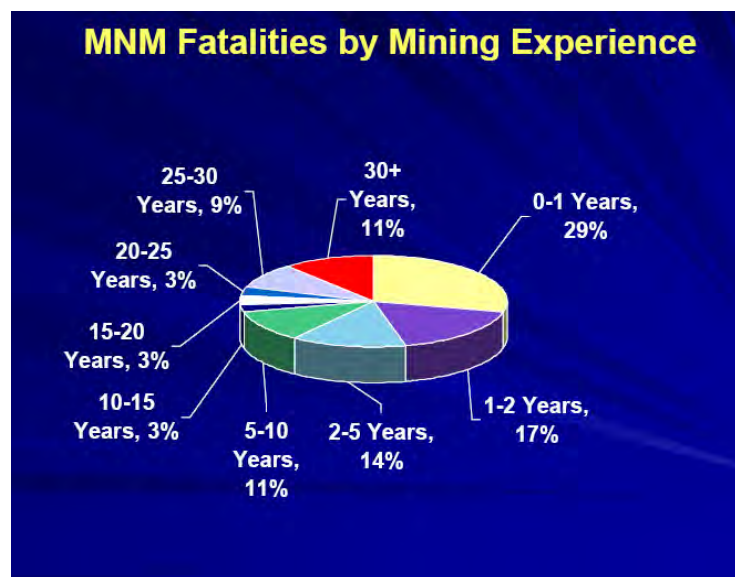
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Figure 4¹Figure 5¹

Root causes of fatalities as shown in Figure 6 demonstrate that inadequate procedures and personal protective equipment (PPE) accounted for the highest percentage of fatalities.

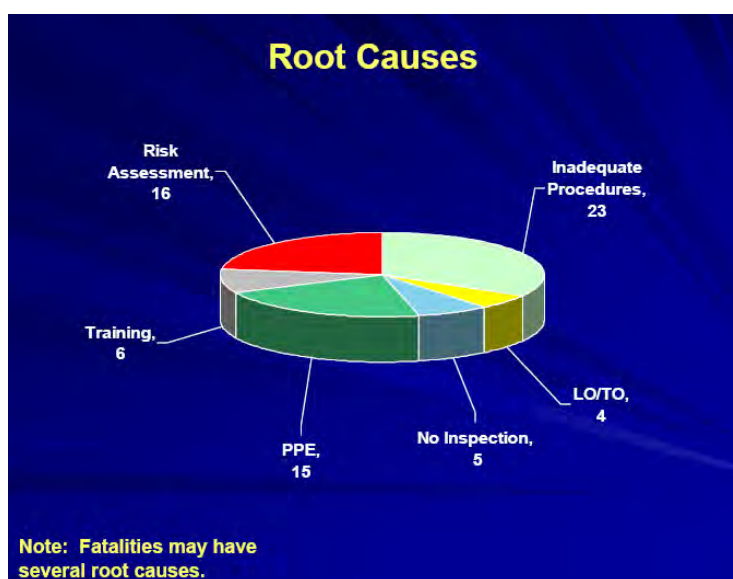
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Figure 6¹

Summarizing the fatality data from 1995 to 2005 provided by MSHA, there are some clear areas where extra attention should be considered, that would include powered haulage units, inexperienced miners, and properly documented and followed safety procedures.

MSHA also reported there were 8,322 nonfatal lost-time injuries reported in the U.S. in 2003. These occurred at a rate of 3.2 injuries per 100 full time equivalent (FTE) workers and resulted in a total of 467,432 days lost from work.² Additional abstracts from that data set include:³

- The underground injury rate was greater than the surface injury rate (6.3 vs. 2.6 per 100 FTE workers).
- The nonfatal lost-time injury rate for mine operator employees was higher than the rate for contractor employees (3.4 vs. 2.0 per 100 FTE workers).
- Coal operator employees had the highest nonfatal lost-time injury rate (4.4 per 100 FTE workers).
- The highest percentage of nonfatal lost-time injuries was due to handling materials (35.4%; n = 2,943) followed by slips or falls (24.9%; n = 2,074).
- The back continued to be the most frequently reported part of the body injured.³ It accounted for 1,755 injuries, 105,510 days lost from work, and 22.6% of all days lost.

Reviewing the fatality data from MSHA for 2005 and non-fatal incidents from 2003, it is clear that additional work is needed to improve safety at mine sites. One method of achieving the objective of zero incidents is the development of well established, followed, and managed safety procedures. This Best Practice will focus on concepts to achieve effective safety procedures.⁴

This Best Practice deals with mine site safety practices and processes that can be used to reduce and minimize safety incidents. Safety incident reporting is sometimes considered taboo and is often the result of a miss understanding of mine management's intent and philosophy. A properly developed safety culture at a mine site would include full support, full reporting, and a vigorous attitude towards implementing and using good safety practices and processes.

This Best Practice will outline high level practices and processes seen in use at a mine sites around the world. It will not provide target metrics or site-specific implementation recommendations as that is entirely up to the reader to consider and implement as needed This Best Practice is meant to make the reader aware of what practices and processes should be in place at a mine site. Ultimately, the key benefit of implementing processes outlined in this Best Practice is for enhancing operator and technician safety.

¹ Does not include underground mining or surface coal mining and MNM – metals/non metals mining.

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² Includes actual days away from work and/or days of restricted work activity. For permanently disabling injuries only, statutory days charged by MSHA were used if they exceeded the total lost workdays.

³ MSHA only reports the most severely injured part of body for accidents involving injury to multiple body parts.

⁴ Data is only shown for the U.S. due to ease of gathering the data, however, empirical evidence supports similar issues throughout the world. All analyses exclude office employees, except for the total number of mining operations, from Mine Safety and Health Administration (MSHA) website, www.msha.gov.

2.0 Best Practice Description

2.1 Philosophy

2.1.1 Safety Vision

A safety vision can be a simple phrase or statement used to summarize an organization's entire underlying safety philosophy. It serves as a daily visual and verbal reminder to all employees that safety is first in everything they do. It helps employees understand that everything done will use will focus on achieving safety. A successful safety program has engagement and true participation from all levels of the mine site. A particular mine's actual safety vision may differ slightly, but the underlying intent is the same. The vision should be written and well communicated throughout the mine site.

2.1.2 Without Harm to People, Property, Environment or Reputation

In support of a safety vision, it must understand that actions to achieve this philosophy must also protect people, property, environment, and reputation.

2.2 Individual Pre-Site Access:

2.2.1 Drug and Alcohol Testing

Drug and alcohol testing is used as part of the employee selection process by assisting in the determination if the potential employee is fit for duty. Drug and alcohol testing is also used post-incident in support of an investigation. An off site, independent lab is used to ensure confidentiality and unbiased results.

2.2.2 Medical Fitness Assessment

A medical physical is given as part of pre-employment analysis to determine if the potential candidate is fit for duty and to provide benchmark data for the employee.

2.3 Personal Commitment

Personal ownership of an employee's responsibility for safety is important. The mine has reported the use of a large poster that outlines the basic safety requirements and is posted in the shop or operations area for all to see. The following statements are included on the poster that all employees voluntarily sign.

- Working safely is a condition of employment when working at the mine
- I believe all accidents are preventable
- I am responsible for my own individual safety performance
- I am responsible for my co-workers safety performance
- I am responsible for the environment, property and reputation of our workplace
- I am responsible to stop unsafe acts noticed

2.4 Orientations and Training

2.4.1 Regulatory Required General Safety Training

Many federal regulations and local governments require the completion of government sponsored mining safety training. In the United States, that includes Mine Safety and Health Administration (MSHA) training, in Alberta, Canada, this includes Construction Safety Training System (CSTS), and in South Australia, that includes Mining and Quarrying Occupational Health and Safety Committee (MAQOHSC) training. This training is conducted prior to arriving on site. All employees would have this training completed on the periodic basis required by the local regulation. Other examples of regulatory required training is occupational health and safety (OH&S).

2.4.2 Regional Orientation

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Due to extremely unique regional hazards and regional safety requirements, a mine site will conduct a regional orientation for all employees. This is a general orientation that focuses on the safety philosophy and general requirements that are unique to that area. Regional orientation would include awareness of culture and climate, fauna, plants and vegetation, and endangered species as examples.

2.4.3 Site Specific Orientation

Prior to admittance of any employee or visitor to a mine site, a mine will provide a site-specific safety orientation course. The training outlines the expectations of all employees and visitors. The site-specific training will include emergency evacuation procedures, emergency contact numbers, emergency alarm types, driving requirements, personal protective equipment requirements, mine property and pit limits, signage in use, hazardous areas of the mine, any regional or governmental training requirements, and local site conditions, as examples.

2.4.4 Area Specific Orientation

Once on site, area specific training is also provided for employees and visitors to the very specific requirements for a particular part of the mine. This may include reporting or permission requirements to conduct work, confined space and welding requirements, or working at heights requirements as examples.

2.4.4.1 Health & Safety, Security, Environmental & Industrial Hygiene

Each department trains their team members in the rules, procedures, standards and risks related specifically to their department. Also included is a review of previous incidents, a department tour, and appropriate personal protective equipment etc.

2.4.4.2 Equipment Operation

In order to operate equipment or vehicles, team members must be trained and certified on each specific vehicle or equipment they need to operate. Training can consist of Computer Based Learning (CBL) specific to the make and model of the equipment, and then checked via evaluation by a specified trainer. This training is required prior to operating all light vehicles, forklifts, skid steers, man lifts, overhead cranes, heavy equipment, etc.

2.4.5 Task Specific Training

This is very similar in scope to Area Specific Training, however it is focused on the very specific tasks used on the job. It is expected that team members be trained to complete all tasks – whether it be how to use the wash bay, how to change oil, how to apply degreaser, or how to remove an engine. This expectation is met by providing experienced workers to train new employees, and using specific procedures established on site involving equipment operation and maintenance.

2.4.6 Hazard Assessment and Recognition Training

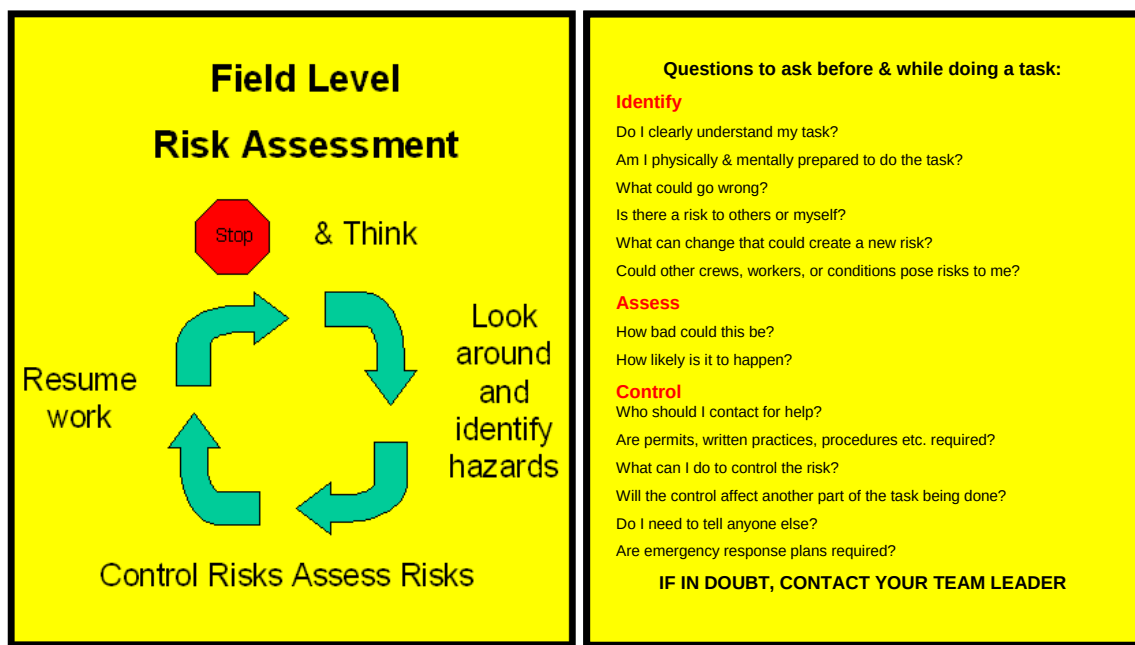
Employees are given a course that teaches workers to identify hazards of each job they are doing, and how to control them. There is a tool (Job Safety Analysis) that is used to document this assessment. These assessments are done for all work where there are hazards or when a Field Level Risk Assessment (FLRA) would not be effective.

2.4.6.1 Field Level Risk Assessments (FLRA)

Workers are trained to stop and evaluate each task, or each part of a task, to identify safety hazards. This assessment is not documented. As an example, an FLRA would be used when a field mechanic was called to diagnose a hydraulic leak on a dozer. A field level risk assessment includes stopping prior to initiating work, looking around and identifying hazards, assess the risks, control the risks, and then resume work.

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2.4.6.2 Job Safety Analysis (JSA)

A formal, documented process used to identify the steps of a job, the potential hazards, and how they will be controlled. This must be done for all work where there is a medium or above risk, and all workers involved in the job must be involved in the JSA and sign it off. For example, a JSA would be required when a worker(s) are assigned to remove and install a front strut on an off-highway truck.

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COMPANY: XYZ		JSA Title: Removing 797 body @ 9800 hrs
JOB BEING ANALYZED: REMOVE DUMP BODY 8-104		JSA REVIEW TEAM
JSA LEADER (NAME): Lead technician		
DATE OF JSA: June 5, 2004		
Nearest Emergency Meeting Point: "D"		
PPE REQUIRED STANDARD PPE		
SEQUENCE OF JOB TASKS	POTENTIAL HAZARDS & EFFECTS	RECOMMENDED PROCEDURES OR CONTROLS
ENSURE CHOCK BLOCKS	ROLLING OF TRUCK	INSTALL BLOCKS
ENSURE ADEQUATE RIGGING	FALLING LOAD	PROPER RIGGING
KEEP WORK AREA CLEAN	TRIPPING HAZARDS	SWEEP UP AS REQUIRED
ENSURE CONNECTED/LOWER EYES	PROPER SHACKLES	EQUIPMENT CLEARANCE
ENSURE WIRING/GREASE LINES ETC	REMOVE	DAMAGE TO EQUIPMENT
ENSURE SAFETY'S ARE ON PINS		DROPPING PINS
ENSURE ADEQUATE VISIBILITY		

Figure 1 Example of a Job Safety Analysis

2.4.6.3 Permit to Work

For high risk tasks (confined space entry, two crane lifts, critical lifts, hot work on tanks, etc) an actual permit is required. This is an extensive document that identifies all possible hazards and how they will be documented. This is developed with the workers involved, the supervisors, a safety person, a member of the emergency response team, and the department manager. The permit is issued by date and time, and cannot run past the end of the shift. All workers participating in the task must review and sign it, and when the work is complete all people involved in the development must close and sign the permit.

2.4.6.4 Planned Inspections/Behavior Based Safety Observations

Hazard recognition training as listed above is required to participate in planned inspections of the work area. This mine site adopted the BEST Behavior Based Safety concept. BEST stands for Building Everyone Safe Tomorrows. All employees without exception are required to be trained in the BEST principles. Additional information on the BEST Behavior Based Safety concept can be found at http://www.y12.doe.gov/news/report/toc.php?vn=1_2

2.4.6.5 Energy Isolation

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Stored energy must be identified and controlled prior to working on a job. This can include electrical, hydraulic, gravitational, pressure, etc. This training and standard ensures workers are able to identify a stored energy source, are able to eliminate the possibility of stored energy release, and are able to test to ensure all stored energy is released.

2.4.7 Due Diligence Training

This course teaches the definition of Due Diligence and also stresses the importance of diligent compliance to rules and procedures. The course focuses on ACTION, that is taking a proactive approach to having due diligence in everything you do.

2.4.8 Hazardous Material Training

Many governments require hazardous material training and would be provided to all employees that work with or have access to hazardous materials.

2.4.9 New Worker Education

This is a short course that identifies some of the safety risks that new workers face, and stresses the importance of asking questions.

2.4.10 Back Injury Prevention

This is a short course focusing on maintaining a health back, awareness, and proper lifting techniques.

2.4.11 Waste Management Guidelines

This training this program focuses on the waste management requirements on site, including waste segregation, storage and disposal.

2.4.12 ISO 14001 Awareness

This program provides a general overview of the ISO 14001 if the mine site has received, and how individuals are accountable for the environmental performance.

2.4.13 Team Leader Training

2.4.14.1 Mine Safety Supervisor Education Program

Training program for supervisors.

2.4.14.2 Risk Assessment Matrix Training

This training provides an in-depth look at how to use a risk assessment matrix and how incidents are rated for actual and potential consequences.

2.4.14.3 Incident Investigation Training

This provides basic investigation tools, focusing on determining immediate and basic causes, and determining corrective actions that are "Action Based."

2.4.14.4 Incident Reporting System Training

All incidents are reported in an electronic incident reporting system. The initial facts must be identified, reported and communicated electronically throughout the site within the shift the incident occurred using a "Flash" report. The investigation must be complete and all action plans completed and documented within 30 days. Only the team leader who initiates the incident report can update, so they own it from start to finish.

2.5 Communications

2.5.1 Asses Meeting Each Shift

At the start of each shift, all employees must attend a safety meeting to determine if their previous shift was safe, and if there are any safety concerns to be raised. This occurs 100% of the time, day, night and weekends – no exceptions.

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2.5.2 Weekly and/or Monthly Safety Meetings

Regular meetings are held for each crew to provide safety education and feedback. Note this is not the time to bring up safety issues – these are to be dealt with immediately.

2.5.3 Daily Statistical Reports

Daily site wide safety report for all incident classifications is provided and posted in common areas for all employees to see. Attached is also a daily learning session/communication related to safety to share with the employees.

2.5.4 Incident Learning Reports

The generic information from all incidents throughout the site is distributed site wide, to communicate incidents and immediate corrective actions taken. The complete details of all significant incidents is circulated in an Incident Learning Package to be reviewed with all teams.

2.6 Injury Management

2.6.1 Focus on First Aid/Near Miss Level

A first aid injury is significant and treated as serious – a worker has been injured and the only thing that has kept this from becoming a more significant injury is luck. That thought process drives the first aid focus. Near miss reporting and investigation is always encouraged through the normal incident reporting system, and through Near Miss Reporting Cards that anonymously report near misses for trending.

2.6.2 Medical Aids

A team leader accompanies a injured worker to medical aid and must remain with the worker until they are released. All medical aids require a conference that involves the most senior site management and involvement from the person involved and their team leader. This formal inquiry is to determine causes and ensure corrective action is firmly in place. There is a time requirement for all medical aid conferences to occur within the time or injury. Typically this time period is within 8 hours of the incident.

2.6.3 Lost Time Incident Reporting Process

A process for how all incidents are reported, responded to, and reviewed after.

2.7 Metric Definitions

The following definitions are provided from a global perspective. Due to the various regulatory agencies around the world that have various reporting requirements, all definitions are provided to assist the reader in gaining a better understanding of the metrics involved in a well managed safety program. A mine site will have to select those metrics that are relevant to that region of the world.

Fatal Injury (F) – An injury that results in death.

Lost Time Injury (LTI) – An injury that results in absence from work. The length of absence required to qualify for an LTI varies from country to country (eg Australia one day, UK three days, South Africa 14 Days)

Severe Injury (SI) – An injury that results in a minimum of two weeks off work. This definition is used by South Africa as the lost time injury measurement.

Fatality Incidence Rate (FIR) – The number of fatalities per set number of employees. This is per 1,000 employees in Australia (sometimes per 100,000 employees is also used). In the US it can refer to fatalities per 200,000 hours worked.

Disabling Injury Frequency Rate (DIFR) – The number of disabling injuries per 200,000 hours worked (US/Canada) or one million hours worked (rest of world). Includes all disabling injuries, whether or not they caused employees to be off work.

Fatal Injury Frequency Rate (FIFR) – The number of fatal injuries per hours worked, normally one per million hours.

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Lost Time Injury Frequency Rate (LTIFR) – The number of lost time injuries per 200,000 hours worked (US/Canada) or one million hours worked (rest of world) – again comparability depends on the LTI definition. Replaced in some country statistics by DIFR.

Duration Rate (DR) – The average time lost for every lost time injury. This is a measure of the severity of the injuries occurring. Calculated by dividing the total number of days lost by the number of lost time injuries.

Severity Rate (SR) – Calendar days lost per 200,000 hours worked (US/Canada) or one million hours worked (rest of world).

No Days Lost (NDL) – Used in US, and refers to all injuries not resulting in a lost work day.

Non Fatal Days Lost (NFDL) – Used in US, and refers to the number of days lost from non fatal injuries.

Reportable/recordable Injury – Used in the US, and includes all incidents that require medical treatment, or result in death, or loss of consciousness, or inability to perform all job duties on any workday after the injury, or temporary assignment or transfer to another job. Injuries involving “first aid only” are not reportable. Other countries define it similarly.

NFDL incidence rate: Use in the US, and is based on NFDL injuries per 200,000 hours worked, MSHA replaced this with an all injury incidence rate in 2003.

All injury incidence rate: Used in the US, and combines disabling injuries, medical – aid injuries and fatalities per 200,000 hours worked. Similar to reportable/recordable frequency rate.

Mining Journal, July 2006, Volume 195 No 1
www.miningjournal.com

3.0 Implementation Steps

Implementation steps are highly dependant on what facets of this Best Practice are desired. Implementation would require identifying what policy or practice is desired, what documentation is needed, what training is required, what communication must be made, and what the follow up and metrics are going to be to ensure proper adaptation of the new policy.

4.0 Benefits

Benefits of a well documented and well accepted/followed safety policy includes reduced number of incidents.

5.0 Resources Required

The resources required to implement this Best Practice are dependant on which part is adopted. The resource list is extensive and not provided in this example.

6.0 Supporting Attachments / References

None.

7.0 Related Best Practices

None.

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8.0 Acknowledgements

The submitting mining company wishes to remain anonymous.

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